

QSearch: Creating marketing opportunities with social media analytics

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Using Google Cloud, QSearch helps clients analyze billions of social media records so that marketers can use the insights to identify and generate new revenue opportunities.



About QSearch

Founded in 2014 and headquartered in Taiwan, QSearch provides products that enable marketers to analyze social networks for product, brand, or topic trends.

Industries: Other

Location: Taiwan

(<https://cloud.google.com/kubernetes-engine/>),
[Cloud CDN](#) (<https://cloud.google.com/cdn/>),

Google Cloud Results

- Records only one error per one million requests for short-URL customer tracking service
- Enables the business to connect to and integrate social media search and measurement tools
- Measures employee productivity and controls access to customer projects

Analysing
billions
in 8 h
days

BigQuery (<https://cloud.google.com/bigquery/>)

App Engine (<https://cloud.google.com/appengine/>)

Compute Engine
(<https://cloud.google.com/compute/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Cloud DNS (<https://cloud.google.com/dns/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Cloud Operations
(<https://cloud.google.com/products/operations>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud Datastore
(<https://cloud.google.com/datastore/>)

Cloud Tasks (<https://cloud.google.com/tasks/>)

Cloud DNS (<https://cloud.google.com/dns/>)

Chrome (<https://chromeenterprise.google/>)

Firebase (<https://firebase.google.com/>)

Cloud APIs (<https://cloud.google.com/apis/>)

TensorFlow (<https://www.tensorflow.org/>)

Google Workspace
(<https://workspace.google.com/>)

Gmail
(<https://workspace.google.com/products/gmail/>)

Google Drive
(<https://workspace.google.com/products/drive/>)

For agencies, businesses, and government organizations, social networks and analytics of user views, behaviors, and demographics. Analyzing social networks for product and topic trends can help marketers identify opportunities, spend campaign dollars wisely, and

manage crises effectively. Taiwan
social media analysis company (QSearch)
providing social network analysis
services to clients since 2014.

Calendar

(<https://workspace.google.com/products/calendar/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

“Most of our clients use our software to plan and
execute campaigns or strategies, explain customer

Cloud IAM (<https://cloud.google.com/iam/>)

Chou, Chief Technology Officer, QSearch. “We
primarily capture data from users of YouTube,
Facebook, and Instagram to help them achieve their
objectives.”

QSearch employs 10 people and offers three
products: Trends, a platform that enables
organizations to observe, analyze, and deploy
influence on social media; Profiling, a tool that
allows organizations to layer social media
information over customer data in customer
relationship management systems, providing a
deeper understanding of audiences; and LinkBundle,
a product that allows publishers to create unique
short URLs and embed retargeting codes so they
can track customer behaviors and interactions.

“Google’s
powerful
information
management
capabilities are

well known and
we are extremely
impressed by
the BigQuery
analytics data
warehouse and
the App Engine
application
development
and hosting
functions.”

—*Shih-En Chou, Chief
Technology Officer,
QSearch*

When establishing the business, Chou established its priorities as aligning infrastructure and data costs with business growth; gaining access to expertise in data capture, analysis, and reporting; meeting client needs for speed, stability, and performance; and launching products quickly.

Cloud is the answer

Chou turned immediately to the cloud to meet these criteria. “Cloud is the only way to launch products and solve business problems as fast as possible,” he says. “On-demand resources enable organizations to build proofs of concept or prototypes, and they do

not have to invest in computing resources before fully verifying business models.”

The business evaluated several cloud services and decided Google Cloud solutions best met its requirements. “Google’s powerful information management capabilities are well known and we are extremely impressed by the BigQuery

(<https://cloud.google.com/bigquery/>) analytics data warehouse and the App Engine

(<https://cloud.google.com/appengine/>) application development and hosting functions,” says Chou.

Between 2014 and 2018, QSearch focused on capturing and analyzing data from social media.

“With BigQuery and visualization tools, we could quickly uncover the insights our business and our customers need,” says Chou. “We can also integrate these seamlessly into our workflows through Google Cloud APIs (<https://cloud.google.com/apis/>).”

Extensive use of Google Cloud

QSearch uses a range of Google Cloud Platform (<https://cloud.google.com/>) and other Google Cloud services to deliver its products and services. These include cloud-based infrastructure as a service through Compute Engine

(<https://cloud.google.com/compute/>), running a customized analytics engine with Apache SOLR enterprise search functionality; orchestration and management of container clusters running

LinkBundle through Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>); Cloud CDN (<https://cloud.google.com/cdn/>) to cache

LinkBundle content close to users using Google's globally distributed edge points of presence; development and hosting of a social analytics engine through App Engine; and BigQuery to process social network data.

“Many of our clients are amazed by the speed of our analytics running on Google Cloud Platform. We help them complete analyses of the interests of customers and prospects in just a few seconds.”

—Shih-En Chou, Chief
Technology Officer,
QSearch

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

provides inter-application messaging and

Stackdriver (<https://cloud.google.com/stackdriver/>)

supports performance monitoring and diagnostics,

while Cloud Storage (<https://cloud.google.com/storage/>)

provides storage and easy retrieval of social data –

including unprocessed data such as posts, fan page

metadata and posted images, and processed social

data in CSV, JSON, or PNG file formats. The

business also operates Cloud SQL

(<https://cloud.google.com/sql/>) and Cloud Datastore

(<https://cloud.google.com/datastore/>) databases to

manage business information and user metadata.

Cloud Tasks (<https://cloud.google.com/tasks/>) enables

QSearch to manage the execution, dispatch, and

delivery of distributed tasks, while Cloud DNS

(<https://cloud.google.com/dns/>) provides scalable

domain name system services.

QSearch also uses Google App Script

(<https://developers.google.com/apps-script/>) and

Chrome Extensions

(<https://chrome.google.com/webstore/unsupported>) to

build on Google Cloud functionality, with App Script

enabling the business to develop a Sheets

(<https://workspace.google.com/products/sheets/>) social

analytics report builder.

Furthermore, QSearch is using Firebase

(<https://firebase.google.com/>) to develop mobile

applications, including a notification service for

popular messaging applications; Custom Search API

(<https://developers.google.com/custom-search/>) to

connect to web and image search capabilities;

YouTube Data API

(<https://developers.google.com/youtube/v3/>) and

YouTube Analytics API

(<https://developers.google.com/youtube/analytics/>) to access, search, and analyze data on the service; and

KnowledgeGraph API

(<https://developers.google.com/knowledge-graph/>) to connect to information sources that complement the Google search engine.

The Google Cloud architecture complements the business' use of the AngularJS JavaScript-based open source web application framework and the TensorFlow (<https://www.tensorflow.org/>) open source software library. QSearch uses these products to measure product awareness by recognizing logos and categories of social pictures.

With Google Cloud, QSearch has developed a social analytics API and database for an ecommerce partner's Key Opinion Leader service, which generates marketing performance reports for merchants.

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—Shih-En Chou, Chief
Technology Officer,
QSearch

The benefits of Google Cloud to QSearch are not limited to providing services to customers. QSearch uses Google Workspace

(<https://workspace.google.com/>) products such as Gmail (<https://workspace.google.com/products/gmail/>), Drive (<https://workspace.google.com/products/drive/>), and Calendar

(<https://workspace.google.com/products/calendar/>) to share ideas, manage customer projects, and collaborate effectively on documents. Cloud Identity & Access Management (<https://cloud.google.com/iam/>) enables the business to build on Google Cloud Platform security features by better controlling permissions to access data captured for client projects.

Using Google Cloud has enabled QSearch to scale its products and services to accommodate new clients as well as heavy demand from existing clients. “Many of our clients are amazed by the

speed of our analytics running on Google Cloud Platform,” says Chou. “We help them complete analyses of the interests of customers and prospects in just a few seconds.”

Billions of records

“The biggest request we had was to analyze 43.2 billion social behavior records,” he adds. “Without BigQuery, we would have needed several days to build a data processing pipeline and process the records. However, BigQuery enabled us to complete this activity in just eight hours and enables our analysts to extract value from the data rather than focus on back-end tasks such as assigning compute resources and stabilizing the network.”

The business has also achieved the stability and performance required by customers for services such as LinkBundle. “We only record one request error for every one million requests, which is amazing,” says Chou. Furthermore, adopting Google Cloud technologies has enabled QSearch to operate a highly scalable business without being forced to recruit more people to meet rising demand. “With Google Cloud, the incremental effort involved in serving new customers is very low,” says Chou.

QSearch is looking to use its understanding of managing engagement with consumers to build a stronger presence across a range of industries. “Our intellectual property and knowledge gives us a robust foundation to embed artificial intelligence into our products,” says Chou. “By working with companies that have a global reach, we could

accelerate the development of artificial-intelligence-powered features such as culturally specific behavior profiling and linguistic services.”